

Data models

Entity: users

Purpose: Stores user accounts for students and administrators of the Enrich system.

Field Name	Data Type	Description
user_id	INTEGER (PK)	Unique user identifier
username	VARCHAR	User's login name
email	VARCHAR	User's email address
password_hash	VARCHAR	Hashed password
role	VARCHAR	Role (student/admin)
created_at	DATETIME	Account creation timestamp

Relationships:

- **One-to-one:** user_settings (user_id)
- **One-to-many:** flashcards (created_by)
- **One-to-many:** saved_flashcards (user_id)
- **One-to-many:** quiz_attempts (user_id)

Entity: user_settings

Purpose: Stores personal interface settings for a specific user.

Field Name	Data Type	Description
user_id	INTEGER (PK, FK)	User identifier
theme	VARCHAR	Theme(Light / Dark).
language	VARCHAR	Language(eng/ua)

Relationships:

- **One-to-one:** users

Entity: flashcards

Purpose: Stores flashcards (both the global database moderated by an admin and custom words created by students).

Field Name	Data Type	Description
flashcard_id	INTEGER (PK)	Unique flashcard identifier
word	VARCHAR	Vocabulary word
difficulty_level	VARCHAR	Difficulty level (e.g. C1)
translation	VARCHAR	Translation of the word.
part_of_speech	VARCHAR	Part of speech (e.g., Adverb).
transcription	VARCHAR	Phonetic transcription
meaning	VARCHAR	Word meaning
example	VARCHAR	Example sentence
created_by	INTEGER (FK)	User who created the flashcard (null for the global database).
created_at	DATETIME	Creation timestamp

Relationships:

- **Many-to-one:** users.
- **One-to-many:** saved_flashcards

Entity: saved_flashcards

Purpose: Tracks which words the user has saved to their personal dictionary (Saved section).

Field Name	Data Type	Description
user_id	INTEGER (FK)	User identifier.
flashcard_id	INTEGER (FK)	Saved flashcard identifier.

Relationships:

- **Many-to-one:** users.
- **Many-to-one:** flashcards.

Entity: quiz_attempts

Purpose: Tracks each attempt of a generated word set (mini-test) for calculating statistics.

Field Name	Data Type	Description
attempt_id	INTEGER (PK)	Unique test attempt identifier.
user_id	INTEGER (FK)	User taking the test.
started_at	DATETIME	Attempt start timestamp.
finished_at	DATETIME	Attempt finish timestamp.
score_percentage	INTEGER	Percentage of correct answers (e.g., 85% correct).

Relationships:

- **Many-to-one:** users.
- **One-to-many:** quiz_answers

Entity: quiz_answers

Purpose: Stores the results for each individual flashcard within a test to track which words were learned and which were not.

Field Name	Data Type	Description
answer_id	INTEGER (PK)	Unique answer identifier.
attempt_id	INTEGER (FK)	Reference to a specific test attempt.
flashcard_id	INTEGER (FK)	The word that was tested.
is_known	BOOLEAN	Result: true (Knew that!) or false (Didn't know).

Relationships:

- **Many-to-one:** quiz_attempts.
- **Many-to-one:** flashcards.